

Q. Write a pseudocode to find the smallest number among three given variables. Implement a decision-making structure to compare the variables.

Start

// Input

// Step 1:

// Initialize three variables a, b and c with given values.

a = Value 1

b = Value 2

c = Value 3

// Step 2: Compare the three variables

IF ( $a < b$  And  $a < c$ ) then

    smallest = a

Else IF ( $b < a$  And  $b < c$ ) then

    smallest = b

Else

    smallest = c

// Step 3: Output the smallest number

Print: "The smallest number is: smallest"

End

Q: Develop pseudocode for a basic calculator that performs multiplication and Division.

// Input : Two numbers (num1, num2) and an operator (op)

// Output : The result of the operation.

Start

// Step 1: Prompt the user for numbers and the operators.

Print "Enter the first number:"

num1 = Input ( )

Print "Enter the Second number:"

num2 = Input ( )

Print "Enter operator (\* / ÷):"

op = Input ( )

// Step 2: Perform calculation based on operator:

If (op == "\*") then

result = num1 \* num2

Else If (op == "/") then

If (num2 == 0) then

Print "Division is undefined"

Else

result = num1 / num2

Else

Print "Invalid operator"

// Step 3: Display the result.

Print "The result is :", result

End,

Q. Write an algorithm to determine whether a number is a prime number.

1, Start

2, Ask the user for a number "n" as input.

3, Check if  $(n < 2)$ , then:

- Print "n is not a prime number".

4, Check if  $(n == 2)$ , then:

- Print "2 is a Prime number".

5, Check if n is divisible by any number from 2 to the square root of "n":

- For "i" from 2 to "sqrt(n)" (square root of "n"):

- if  $(n \% i == 0)$ , then:

- Print "n is not a Prime number because it is divisible by i".

- Else

- Print "n is a Prime number because there is no divisor found".

6, End.

Q: Develop an algorithm for a program that takes two numbers as input and find the GCD of two num using Euclidean algorithm.

1. Start
2. Input the two numbers "a" and "b":
  - read 'a'
  - read 'b'
3. Repeat until b becomes 0.
  - Calculate the remainder of "a divided by b" and store it in a variable 'r'.
    - $r = a \% b$
  - Assign value of "b to a".
    - $a = b$
  - Assign value of "r to b".
    - $b = r$
4. When b becomes 0, "a" will contain the GCD:
  - $gcd = a$
5. Output the GCD.
  - Print "The GCD of the two number is", gcd.
6. End.